FMNH Rapid Inventory Use Case



Introduction

The Field Museum & Rapid Inventories (RIs)

Keller Science Action Center

We are dedicated putting science into action that creates and supports lasting conservation that benefits local people




27 YEARS
34 INVENTORIES
31.5 MILLION ACRES CONSERVED

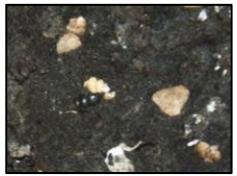
Rapid Inventories

Quick, cooperative, rigorous surveys of the biological and social assets of a priority landscape for conservation





Data Types and Structure



Geology



Vegetation and flora



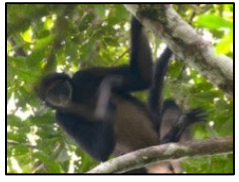
Fishes



Amphibians and reptiles



Birds



Mammals



Social organization and assets



Local ecological knowledge



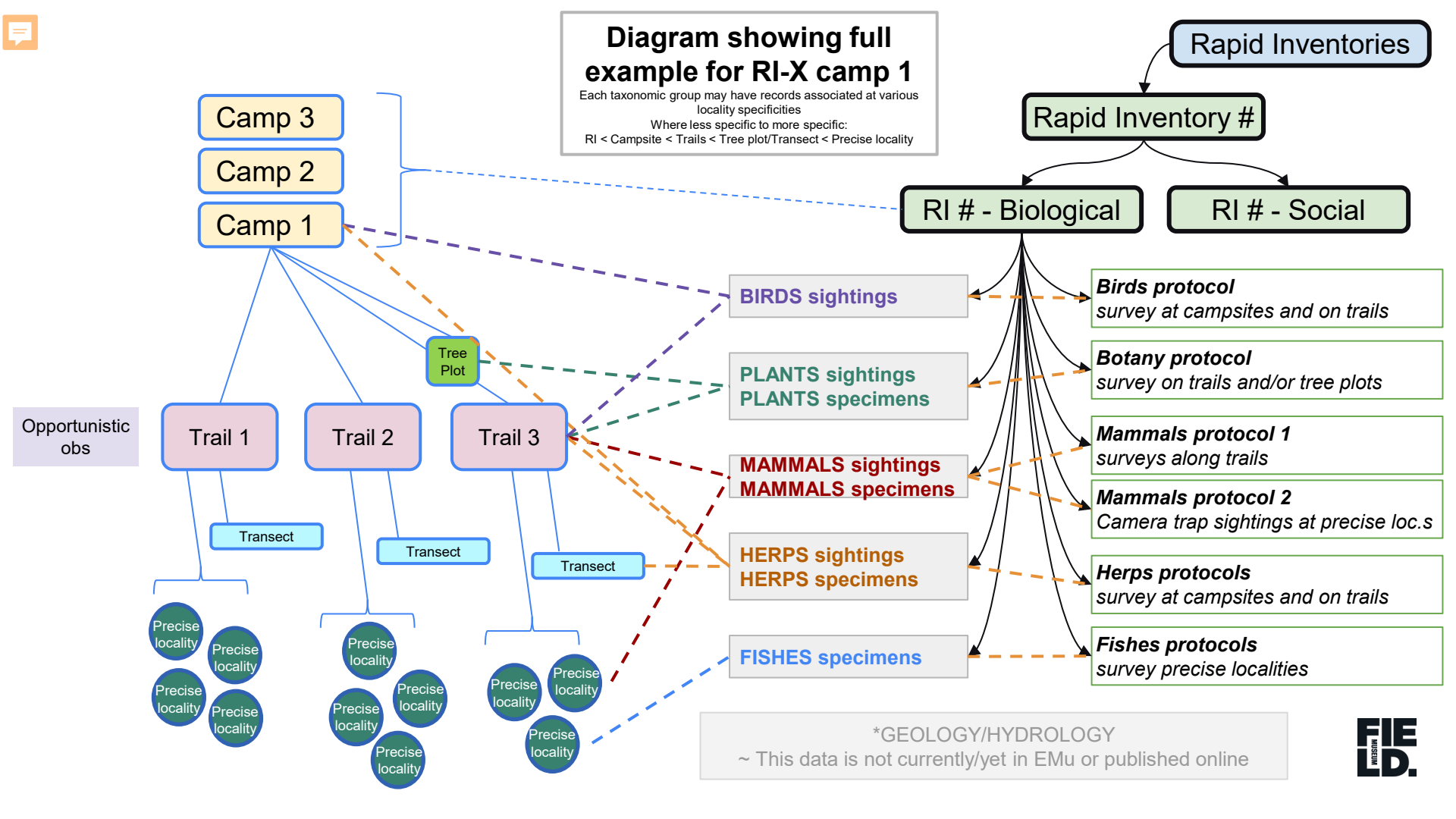
Culture and traditions



History and aspirations



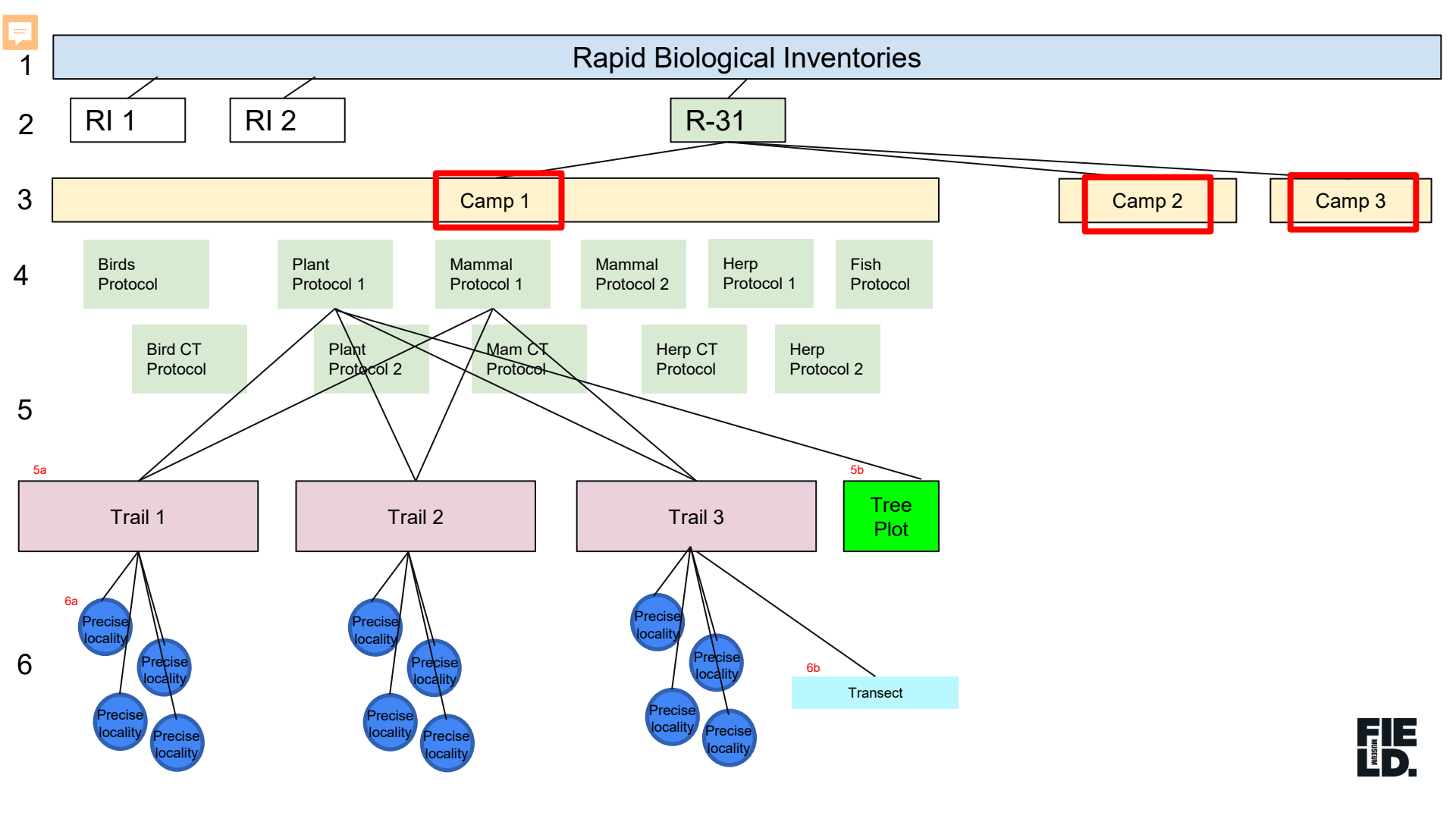
Threats

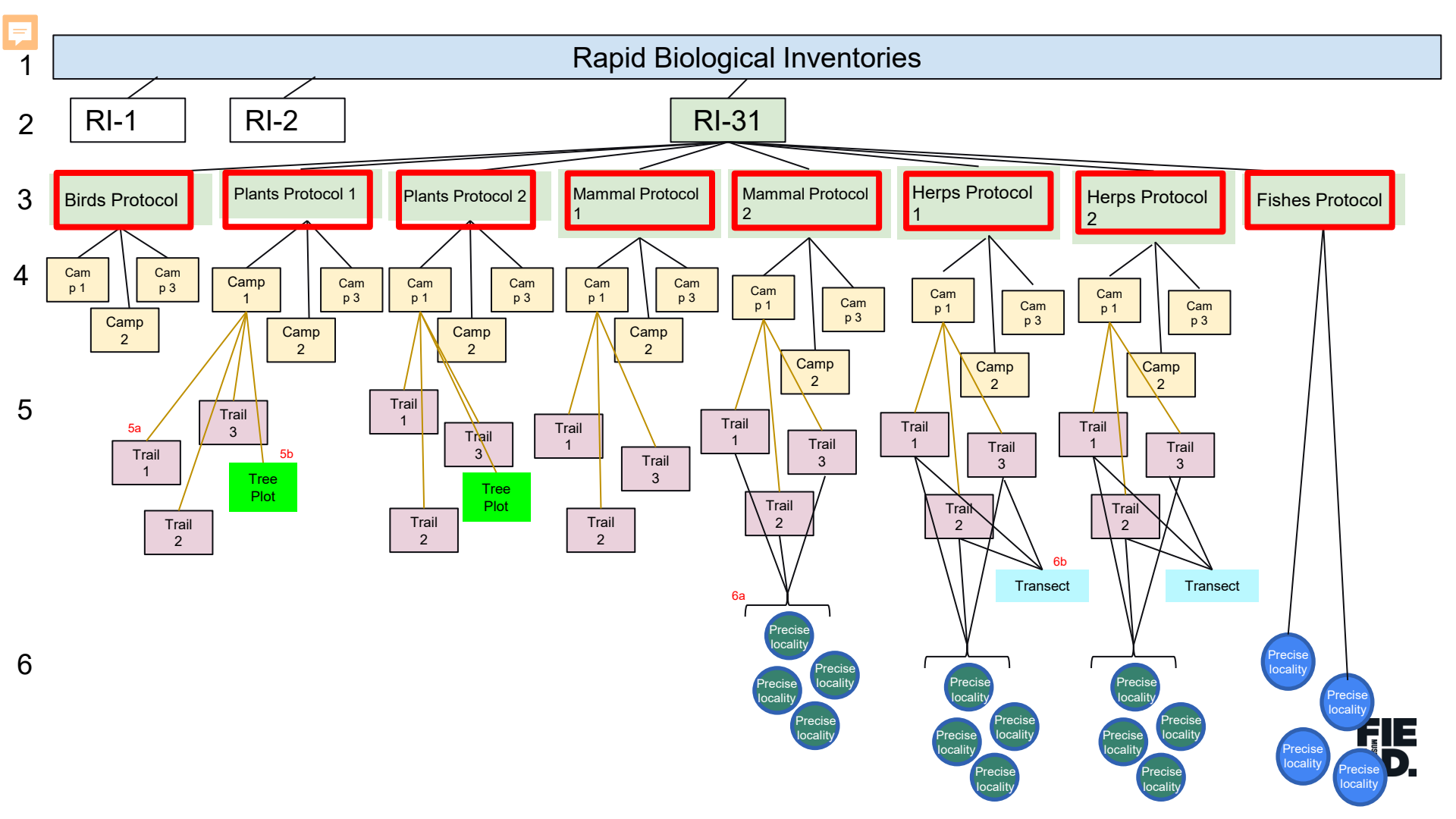

Diagram showing full example for RI-X camp 1
Each taxonomic group may have records associated at various locality specificities
Where less specific to more specific:
RI < Campsite < Trails < Tree plot/Transect < Precise locality
Rapid Inventories
Rapid Inventory #
RI # - Biological
RI # - Social
Camp 3
Camp 2
Camp 1
Tree Plot
Opportunistic obs
Trail 1
Trail 2
Trail 3
Transect
Transect
Transect
Precise locality
BIRDS sightings
PLANTS sightings
PLANTS specimens
MAMMALS sightings
MAMMALS specimens
HERPS sightings
HERPS specimens
FISHES specimens
Birds protocol
survey at campsites and on trails
Botany protocol
survey on trails and/or tree plots
Mammals protocol 1
surveys along trails
Mammals protocol 2
Camera trap sightings at precise loc.s
Herps protocols
survey at campsites and on trails
Fishes protocols
survey precise localities
*GEOLOGY/HYDROLOGY
~ This data is not currently/yet in EMu or published online

Two Ways to Represent the Event + Parent Event Structure

Spatial (location) defined events

Taxonomic (scope/protocol) defined events


1
Rapid Biological Inventories
2
RI 1
RI 2
R-31
3
Camp 1
Camp 2
Camp 3
4
Birds Protocol
Plant Protocol 1
Mammal Protocol 1
Mammal Protocol 2
Herp Protocol 1
Fish Protocol
Bird CT Protocol
Plant Protocol 2
Mam CT Protocol
Herp CT Protocol
Herp Protocol 2
5
5a
5b
Trail 1
Trail 2
Trail 3
Tree Plot
6
6a
Precise locality
6b
Transect

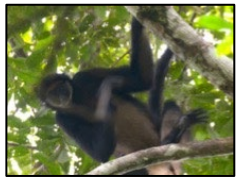

1
Rapid Biological Inventories
2
RI-1
RI-2
RI-31
3
Birds Protocol
Plants Protocol 1
Plants Protocol 2
Mammal Protocol 1
Mammal Protocol 2
Herps Protocol 1
Herps Protocol 2
Fishes Protocol
4
Cam p 1
Cam p 3
Camp 2
Camp 1
5
5a
5b
Trail 1
Trail 2
Trail 3
Tree Plot
6a
6b
Transect
6
Precise locality
FIELD.

The Challenge

CONCEPTUAL MODEL



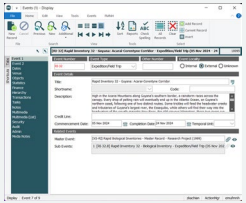


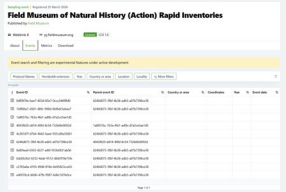


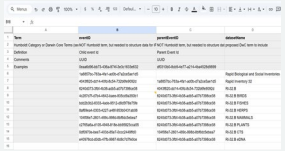




PHYSICAL MODEL







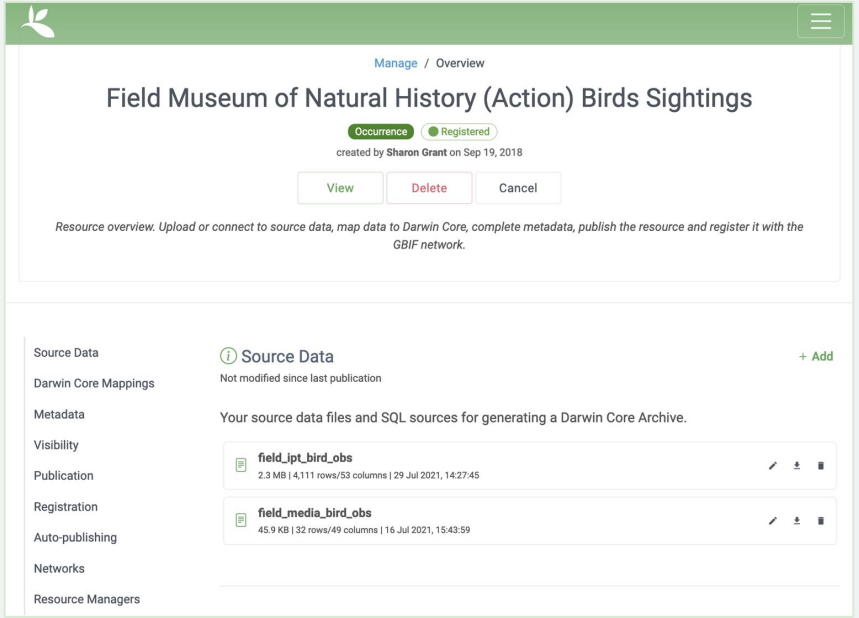
Sampling Event + Humboldt Extension

FMNH IPT resources

1 DwC occurrence dataset per FMNH Collection

Plus extensions:

- Audiovisual Core *(current)*
  - camtrap-dp (future)
- Related Resource
  (currently in specimen datasets)
- Measurement or Fact *(future)*



LOCATION-BASED EVENTS

eventID	parentEventID
Rapid Inventories Program	
Rapid Inventory 32	Rapid Inventories Program
RI32-C1	Rapid Inventory 32
RI32-C2	Rapid Inventory 32
RI32-C3	Rapid Inventory 32
RI32-C1-T1	RI32-C1
RI32-C1-T2	RI32-C1
RI32-C1-T3	RI32-C1
RI32-C1-T4	RI32-C1
RI32-C2-T1	RI32-C2
RI32-C2-T2	RI32-C2
RI32-C2-T3	RI32-C2
RI32-C3-T1	RI32-C3
RI32-C3-T2	RI32-C3
RI32-C3-T3	RI32-C3
RI32-C3-T4	RI32-C3
RI32-C1-Plot	RI32-C1
RI32-C2-Plot	RI32-C2
RI32-C3-Plot	RI32-C3

targetTaxonomicScope: Aves | Mammalia | Reptilia | Amphibia | Actinopterygii | Embryophyta

protocolDescriptions: Birds Protocol | Mammals Protocol 1 | Mammals Protocol 2 | Plants Protocol 1 | Plants Protocol 2 | Herps Protocol | Fishes Protocol

LOCATION-PROTOCOL HYBRID EVENTS

eventID	parentEventID
B	C
RI31Birds	RI31
RI31B1Birds	RI31Birds
RI31B2Birds	RI31Birds
RI31B3Birds	RI31Birds
RI31B4Birds	RI31Birds
RI31Mammals	RI31
RI31B1Mammals	RI31Mammals
RI31B2Mammals	RI31Mammals
RI31B3Mammals	RI31Mammals
RI31B4Mammals	RI31Mammals
RI31B1T1Mammals	RI31B1Mammals
RI31B1T2Mammals	RI31B1Mammals
RI31B1T3Mammals	RI31B1Mammals
RI31B2T1Mammals	RI31B2Mammals
RI31B2T2Mammals	RI31B2Mammals
RI31B2T3Mammals	RI31B2Mammals
RI31B2T4Mammals	RI31B2Mammals
RI31B3T1Mammals	RI31B3Mammals
RI31B3T2Mammals	RI31B3Mammals
RI31B3T3Mammals	RI31B3Mammals
RI31B4T1Mammals	RI31B4Mammals
RI31B4T2Mammals	RI31B4Mammals
RI31B4T3Mammals	RI31B4Mammals
RI31B4T4Mammals	RI31B4Mammals
RI31Herps	RI31
RI31B1Herps	RI31Herps
RI31B2Herps	RI31Herps



A really long, complicated list of location + protocol events

Sampling event | Registered 25 March 2026

Field Museum of Natural History (Action) Rapid Inventories

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About | Events | Metrics | Download

Event search and filtering are experimental features under active development

Protocol Names | Humboldt extension | Year | Country or area | Location | Locality | More filters

10 results

Event ID	Parent event ID	Country or area	Coordinates	Year	Event date
0df0970e-bee7-403d-85e7-0ccc2449ffd0	6240d073-3fbf-4b38-adb5-a07b7398ce38	Camera traps			
104f06e7-2601-489c-998d-8bf6dc5ebea7	6240d073-3fbf-4b38-adb5-a07b7398ce38	Mammals			
1a8857bc-763a-4fa1-ad0b-d7a2ce5ae1d5		Rapid Inventories Program			
4043f620-dd14-40fd-8c54-732b6fe9092d	1a8857bc-763a-4fa1-ad0b-d7a2ce5ae1d5	Rapid Inventory 32			
4c287d7f-d7b4-4642-baee-935cd9a350b1	6240d073-3fbf-4b38-adb5-a07b7398ce38	Birds			
6240d073-3fbf-4b38-adb5-a07b7398ce38	4043f620-dd14-40fd-8c54-732b6fe9092d	Rapid Inventory 32 - Biological Surveys			
8af69ea4-0303-4227-a46f-953b0431ab06	6240d073-3fbf-4b38-adb5-a07b7398ce38	Amphibians & Reptiles			
bdd2b3b2-6333-4ade-9512-d8d5f78e75fe	6240d073-3fbf-4b38-adb5-a07b7398ce38	Fishes			
c2765a6a-d105-4948-818e-bb95823cca55	6240d073-3fbf-4b38-adb5-a07b7398ce38	Plants			
e40978cd-d0db-47fb-9987-6d8c7d7fe0ce	6240d073-3fbf-4b38-adb5-a07b7398ce38	eDNA			

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Thank you!

Questions / Crises?

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